Initial Client Meeting: Nova Haven — UrbanPulse Analytics

Date: March 18, 2026
Attendees:

- City of Nova Haven — UrbanPulse Operations Leadership Team
- AI Consulting Group (1 of 2 finalists)

You are competing against another world-class AI consulting firm for a landmark contract worth $2.8M initially, with $12M+ in guaranteed follow-on work. As the most ambitious smart city initiative in the country, Nova Haven only partners with the best. The winning team will become our exclusive AI innovation partner and help shape the future of urban operations.

About Nova Haven & UrbanPulse Analytics

Nova Haven is a major metropolitan city of 8.3 million residents. UrbanPulse is Nova Haven's smart city analytics division, created by the Mayor's Office of Technology & Innovation to transform how the city operates. Launched in 2024, we're building the most advanced urban AI platform in the country — one that uses machine learning, computer vision, and natural language processing to make city services faster, safer, and more responsive.

Our Mission & Scale

- **Population Served:** 8.3 million residents
- **Annual 311 Complaints:** 3.4 million service requests per year
- **Road Network:** 19,000+ lane-miles of roadway
- **Traffic Incidents:** 250,000+ reported accidents annually in the metro area
- **Pothole Repairs:** 500,000+ potholes filled per year ($100M+ annual cost)
- **Budget Impact:** Every 1% improvement in city operations saves $15M annually
- **Vision:** Make Nova Haven the safest, most responsive city in the world by 2030

The Challenge: Smarter City Operations

Our city generates enormous amounts of data every day — traffic patterns, road conditions, resident complaints, infrastructure reports. But we're drowning in data and starving for intelligence. We need AI to turn this data into action.

Current Urban Challenges

- **Traffic Safety:** 250,000+ accidents/year, severity prediction is essentially nonexistent
- **Road Maintenance:** Reactive, not proactive — we fix potholes after residents complain
- **311 Response:** Average 3.2 days to route complaints to the right agency
- **Knowledge Access:** City employees spend 20+ minutes searching for policy answers
- **Resource Allocation:** Snow plows, repair crews, inspectors deployed by gut feel, not data

Our Pain Points

1. **Traffic Accident Severity**
    - No predictive capability — we respond to accidents, we don't prevent them
    - Can't prioritize intersection improvements based on data
    - Weather, road features, and time patterns are collected but not leveraged
2. **Road Surface Conditions**
    - $100M+ annual pothole repair budget with no prioritization intelligence
    - Inspectors manually survey roads — slow, expensive, inconsistent
    - Citizen complaints are the primary detection method (too late)

3. **311 Complaint Processing**
    - 186 complaint categories — agents take 4+ minutes to classify each call
    - Misrouted complaints add 2-5 days to resolution time
    - Complaint text contains rich information that's never analyzed

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Available Data Assets

> ⚠️ **IMPORTANT: Read the Preprocessing Hints File**
> The `preprocessing_hints.py` file contains data preparation guidance that will significantly impact your model performance. Review it carefully before starting any modeling work.

1. City Traffic Accident Database

File: city_traffic_accidents.csv (~500,000 accident records)

Comprehensive Accident Records Including:

Severity:

- Severity rating (1-4 scale) measuring **traffic disruption**, NOT injury severity:
    - **1** = Short delay, minimal traffic impact
    - **2** = Moderate delay
    - **3** = Long delay, significant traffic impact
    - **4** = Major delay, severe traffic disruption (e.g., multi-lane highway closure)
- A fender-bender blocking a highway on-ramp could be Severity 4, while a serious injury on a rural road might be Severity 1. **You are predicting how much traffic was disrupted, not how badly anyone was hurt.**
- This dataset has significant class imbalance — explore the severity distribution carefully during EDA.

Temporal:

- `Start_Time` / `End_Time`: When the accident occurred and when traffic impact ended

- `Sunrise_Sunset`: Whether it was day or night
- Twilight conditions (civil, nautical, astronomical)

Location:

- `Start_Lat` / `Start_Lng`: Where the accident occurred
- `End_Lat` / `End_Lng`: Where the **traffic impact ended** (e.g., end of backup)
- `Distance(mi)`: Length of **road affected** by the accident, NOT distance traveled
- City, zip code, street name, side of road

Weather Conditions:

- Temperature, humidity, wind speed/direction, visibility, precipitation, weather condition text, wind chill, pressure
- Missing weather values are NOT random — they mean the nearest weather station was offline or the API didn't return data for that location/time

Road Features (boolean flags):

- Multiple boolean flags indicating the **presence of nearby infrastructure** at the accident location — NOT characteristics of the accident itself

Description:

- Free-text description of the accident

Data Collection Note: This accident data was collected via traffic monitoring APIs that aggregate reports from transportation departments, law enforcement, traffic cameras, and road sensors. Keep collection bias in mind when interpreting geographic patterns.

2. Road Surface Images

Directory: pothole_images/ (~4,400 road surface images)

Computer Vision Dataset:

- High-resolution road surface photographs
- Binary classification: pothole vs. normal road surface
- Images organized in `positive/` (pothole) and `negative/` (normal road) folders

- Collected via dashboard camera
- Mix of road types, lighting conditions, weather conditions
- The dataset has class imbalance — explore the distribution during EDA

Note: Images are very high resolution — you will need to resize for training.

3. Nova Haven 311 Service Requests

File: urbanpulse_311_complaints.csv (~500,000 recent complaint records)

Citizen Complaint Records:

- Unique request ID and timestamps (created, closed)
- Agency and agency name handling the request
- **Complaint type** (target for classification)
- **Resolution description** (free text — how the complaint was resolved)
- Location data (zip code, district, lat/lng, address)
- Status (open, closed, pending)
- Channel (phone, online, mobile, other)

For your NLP model: Classify complaints into the correct `complaint_type` category using the `resolution_description` text field. There are many complaint categories, but we recommend focusing on the **top 5 categories plus an "Other" bucket** (6 classes total): Illegal Parking, HEAT/HOT WATER, Noise - Residential, Snow or Ice, and Blocked Driveway. This is a practical real-world decision that city operations teams would actually make.

What We Need From You

5 Required AI Models

Model 1: Traffic Accident Severity — Traditional ML

- Multi-class classification: predict accident severity (1-4 scale)

- This dataset has significant class imbalance — your approach to handling it will be a key evaluation criterion. **Weighted F1 is the real evaluation metric**, not accuracy.
- Use classical ML algorithms (XGBoost, Random Forest, Gradient Boosting, etc.)
- Must be interpretable — city planners need to understand WHY an intersection is flagged as high-risk
- **Minimum Benchmark:** Accuracy > 70%, weighted F1 > 0.55
- **Stretch Goal:** Accuracy > 80%, weighted F1 > 0.70
- **Required:** SHAP or feature importance analysis — which factors most predict severe accidents?

Model 2: Traffic Accident Severity — Deep Neural Network

- Same prediction task as Model 1, but using a deep learning approach
- The same class imbalance challenge applies — evaluate with weighted F1
- Build a neural network (TensorFlow/Keras) on the tabular accident data
- Compare performance against your traditional ML model
- **Minimum Benchmark:** Accuracy > 70%, weighted F1 > 0.55
- **Stretch Goal:** Accuracy > 82%, weighted F1 > 0.72
- **Required:** Show how DNN performance compares to traditional ML and explain the trade-offs

Model 3: Pothole Detection — CNN

- **Binary image classification:** pothole vs. normal road surface
- The dataset has class imbalance — a model predicting the majority class for everything is useless
- Build or fine-tune a convolutional neural network
- **Minimum Benchmark:** Accuracy > 82%, weighted F1 > 0.78
- **Stretch Goal:** Accuracy > 92%, weighted F1 > 0.90
- **Also valued:** Grad-CAM visualizations showing what the model "sees"

Model 4: 311 Complaint Classification — NLP

- Multi-class text classification: predict the complaint type from text
- **Top 5 categories + "Other" (6 classes total):** Illegal Parking, HEAT/HOT WATER, Noise - Residential, Snow or Ice, Blocked Driveway, and Other (everything else)
- Use NLP techniques (TF-IDF + classifier, embeddings, or transformers)
- Handle real-world citizen text (informal language, abbreviations, multiple languages)
- **Minimum Benchmark:** Accuracy > 75%, weighted F1 > 0.70
- **Stretch Goal:** Accuracy > 85%, weighted F1 > 0.80
- **Also valued:** Complaint routing recommendations

Model 5: Urban Innovation — Your Choice

- Identify an urban operations problem in the data that we haven't asked you to solve
- Build a model that addresses a genuine city operations need
- **You must use the data we've provided** — do not bring in external datasets. The goal is to show us what's possible with what we already have.
- Ideas to explore: accident hotspot prediction, seasonal complaint patterns, response time optimization, road deterioration prediction, resource deployment optimization
- Think creatively about how to leverage our data in ways we haven't considered — could you use agents, RAG, or other innovative techniques to extract more value?
- **Must include:** Clear urban value proposition, defined success metric, cost-benefit estimate
- **We want you to surprise us.** The winning team won't just build what we asked for — they'll show us opportunities we didn't see.

Project Timeline (~7 Weeks)

Phase 1: Data Foundation & Discovery (Weeks 1-2)

Week 1 — Data Audit & Exploration

- Download and load all datasets
- Comprehensive data quality assessment
- Initial EDA — what patterns exist in accidents, complaints, road conditions?

Week 2 — Feature Engineering & Baselines

- Build your data pipeline (`pipelines/data_pipeline.py`)
- Feature engineering across all datasets
- Establish baseline models for Models 1 & 2
- Begin image and text preprocessing for Models 3 & 4

Phase 2: Model Development (Weeks 3-5)

Week 3 — Core Models

- Model 1 (Traditional ML): Iterate on algorithms, feature selection, handle class imbalance
- Model 2 (DNN): Build neural network, compare to Model 1
- Model 3 (CNN): Build/fine-tune image classification on pothole data
- Model 4 (NLP): Build text classification for 311 complaints

Week 4 — Advanced Models & Integration

- Model 5 (Innovation): Develop your chosen urban innovation
- Begin web app development
- Start integrating models into the application

Week 5 — Optimization & Testing

- Tune all models for best performance
- Ensure output formats match templates exactly
- Run bulk test scripts — fix any issues
- Test on validation data — check for overfitting and data leakage

Phase 3: Deployment & Presentation (Weeks 6-7)

Week 6 — Integration & Deployment

- Complete web application with all 5 models integrated
- Deploy to hosting platform
- End-to-end testing
- Prepare presentation materials

Week 7 — Presentation Prep & Final Testing

- Polish presentation deck
- Practice live demo
- Prepare for Q&A
- Final submission and code cleanup

Sprint Reviews

- **Weekly check-ins** with your instructor (acting as the Nova Haven UrbanPulse technical lead)
- **Submit your** `weekly-sprint-template.md` **via Slack before each check-in** — this is required
- Come prepared with: progress update, blockers, questions, demo of current state
- These are working sessions — bring your laptop and be ready to show code

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Success Metrics & Urban Impact KPIs

Primary Outcomes

From Required Models:

- **Accident Prediction:** Enable proactive safety interventions at high-risk intersections
- **Pothole Detection:** Shift from reactive to proactive road maintenance
- **311 Processing:** Reduce complaint routing time from 3.2 days to < 1 day

Model Evaluation Framework

Models are evaluated holistically, not just on accuracy:

Urban Impact (35%)

- Meets minimum performance benchmarks
- Predictions are actionable for city operations
- Validated on realistic scenarios

Reliability & Fairness (25%)

- No harmful biases across districts or demographics
- Handles edge cases and missing data gracefully
- Clear limitations documented

Usability (25%)

- Intuitive interface for city employees (not data scientists)
- Interpretable for city planners (SHAP/LIME/Grad-CAM)
- Integrates into existing workflows
- Response time < 500ms for prediction models
- Clear confidence indicators

Scalability (15%)

- Can handle city-scale data volumes
- Cost-effective deployment
- Extensible to other cities

How We'll Evaluate You

Technical Deliverables (45% of grade)

Component	Points	What We're Looking For
Model 1: Traditional ML	18	Performance metrics + interpretability (SHAP/feature importance)
Model 2: Deep Learning	18	DNN architecture + comparison to Model 1
Model 3: CNN	18	Image classification accuracy + handling imbalance
Model 4: NLP	18	Text classification + handling real-world text
Model 5: Innovation	18	Urban value + technical quality

Critical — How Testing Works:

Your models will be tested on **completely unseen data** that we hold back. This test data looks exactly like the raw data you downloaded — same columns, same formats, same messiness (e.g., missing weather values, mixed text formatting, varied image conditions). Your `predict.py` must:

1. **Accept raw data** in the same format as what you downloaded (not pre-cleaned data)
2. **Apply all preprocessing automatically** — the same cleaning, encoding, and feature engineering you did during training must happen inside your prediction pipeline
3. **Output predictions** in the exact format specified by the output templates

Think of it this way: we will drop a fresh CSV (or images) into your `test_data/` folder and run `python bulk_test.py --all`. Your code needs to handle everything from raw input to final predictions **without any manual steps**. If your pipeline only works on data you've already cleaned by hand, it will break.

Build your preprocessing as reusable functions in `pipelines/data_pipeline.py` and import them in both your `train.py` and `predict.py`. This is how real production ML works — the same pipeline handles training data and new data.

Business Deliverables & Presentation (45% of grade)

- Professional slide deck presenting your solution and urban impact case
- Live demo of your deployed web application
- Q&A performance — ability to answer technical, operational, and strategic questions
- Clear justification of your decisions and feature prioritization
- Professional communication and presentation skills

Collaboration & Process (10% of grade)

- Equal participation and contribution from all team members
- Effective teamwork and knowledge sharing
- Professional project management and communication
- Peer evaluations will influence individual scores

Evaluation Questions for All Competing Firms

Each firm will be scored on these responses during the final presentation:

1. How will you handle missing weather data in accident prediction?
2. What's your approach to the severe class imbalance in accident severity?
3. How will you ensure your pothole detector works across different road types and lighting?
4. What data augmentation strategies did you use for image classification?
5. How did you select the top 5 complaint categories and handle the "Other" bucket?
6. How does your deep learning model compare to your traditional ML model, and why?
7. How would your models adapt to a different city?
8. What's your approach to model interpretability for city planners and operations staff?
9. How will you handle real-time data streams in production?

10. **Differentiator Question:** What innovative approach will you bring that your competitors won't?

Data Quality & Integration Notes

Real-World Data Challenges:

- Accident severity is based on traffic impact, not injury — understand what you're predicting
- Class imbalance exists across datasets — strategies to handle it are essential
- Weather columns have missing values — these aren't random
- Pothole images are very high resolution — resize before training
- 311 text data includes multiple languages — plan for this
- Some accident descriptions are auto-generated and may follow templates

This messiness is intentional — real city data is never perfect. Your ability to handle these challenges is part of the evaluation.

Common Pitfalls (Learn from Previous Teams)

Teams have failed because they:

- Didn't standardize their output format (columns didn't match, evaluation script failed)
- Left final code in notebooks instead of `.py` files
- Ignored class imbalance and reported misleading accuracy numbers
- Built models with high accuracy but poor interpretability — city planners couldn't understand or trust the predictions
- Didn't test their bulk_test scripts before submission

Teams have succeeded because they:

- Followed the repo template exactly
- Tested their output format against the templates early

- Made thoughtful decisions about which complaint categories matter most
- Told a compelling story about urban impact in their presentation
- Showed genuine understanding of city operations

Professional Communication Expectations

As our potential smart city AI partner, we expect consultant-level professionalism:

- **Think like a city planner, not just an engineer —** understand why these problems matter to residents
- **Frame everything in terms of public impact** — lives saved, response times improved, taxpayer dollars saved
- **Never criticize our data quality —** we work with real-world city data
- **Speak like a partner, not a vendor** — you're helping us transform our city
- **Remember:** Behind every 311 complaint is a resident who needs help

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Contract Competition Details

- **Evaluation Period:** ~7 weeks to prove your capabilities
- **Weekly Milestones:** Sprint reviews each week
- **Final Presentations:** May 13 & May 14, 2026 (both firms present)
- **Winner Selection:** May 15, 2026

Important Notes

1. **Open Data:** All datasets have no privacy concerns
2. **Urban Context:** Your instructor serves as the Nova Haven UrbanPulse operations team — ask questions!
3. **Scalability:** Solutions should be designed to handle city-scale data volumes
4. **Real Impact:** Your models would directly affect how Nova Haven serves 8.3 million residents

"Every minute a 311 complaint is misrouted is a minute a resident waits for help. Every pothole we miss is a bike accident waiting to happen. Every accident we can't predict